



• 100% FREE · Live Cohort

👋 Hey beginners — this one's for you

Get started with Cloud & AI on AWS in just 5 days.

A friendly, **beginner-friendly intro** to AWS — no prior experience needed. Perfect for students, graduates, and anyone curious about cloud or AI.

- ✓ **Learn the essentials** — cloud, storage, compute, AI services
- ✓ **Hands-on demo on Day 5** — see AWS services working together
- ✓ **10 hours total** · live · 3 timezones to choose from

★★★★★

"Concepts are explained in a simple, step-by-step manner. Even advanced AWS topics feel clear and manageable."

Gururaj Rathod · DevOps Engineer

★★★★★

"I still go back to it as a reference. So much content — theory plus hands-on with real-world examples."

Antonio Villarán López · Cloud Architect

LIVE DATES

May 25 – 29

Mon to Fri · 2026 · 2 hrs/day

PICK YOUR TIMEZONE

EST 9:00 – 11:00 AM

GMT 2:00 – 4:00 PM

IST 6:30 – 8:30 PM

Save your seat at

iaasacademy.com

→ Limited live spots



FOR BEGINNERS · NO EXPERIENCE NEEDED

CLOUD & AI ON AWS

100%
FREE
5-DAY COHORT



DAY 4 · A Gentle Intro to **AWS AI**

MAY 25-29 · 2 HRS/DAY · **LIVE**

Agenda

Day 1 ● What is cloud computing, AWS and Identity & access (IAM)



Day 2 ● Introduction to Storage (Amazon S3), and content delivery (AWS CloudFront)



Day 3 ● Compute (EC2, Lambda, ECS) and how services communicate (SQS, SNS)



Day 4 ● A friendly intro to AWS AI services: Bedrock, Rekognition, Polly, Textract

Day 5 ● Hands-on demo: Lambda, Textract, Bedrock, S3 and CloudFront chained together



Introduction to AI/ML Services on AWS

High Level Introduction – AI tools to build solutions

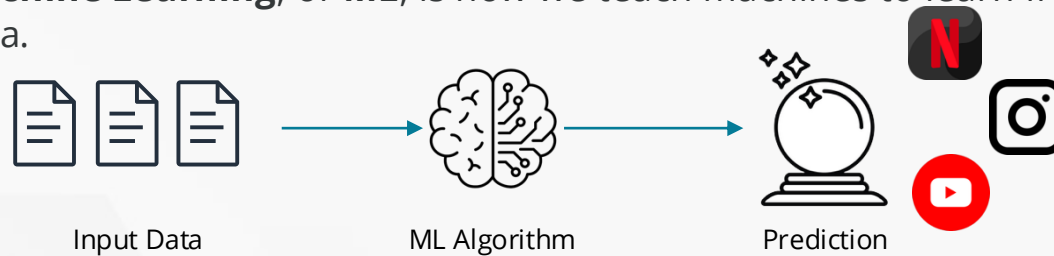
Introduction to Artificial Intelligence

Artificial Intelligence

- Creating machines that can perform tasks that normally require human intelligence

Machine Learning

- **Machine Learning**, or **ML**, is how we teach machines to learn from data.



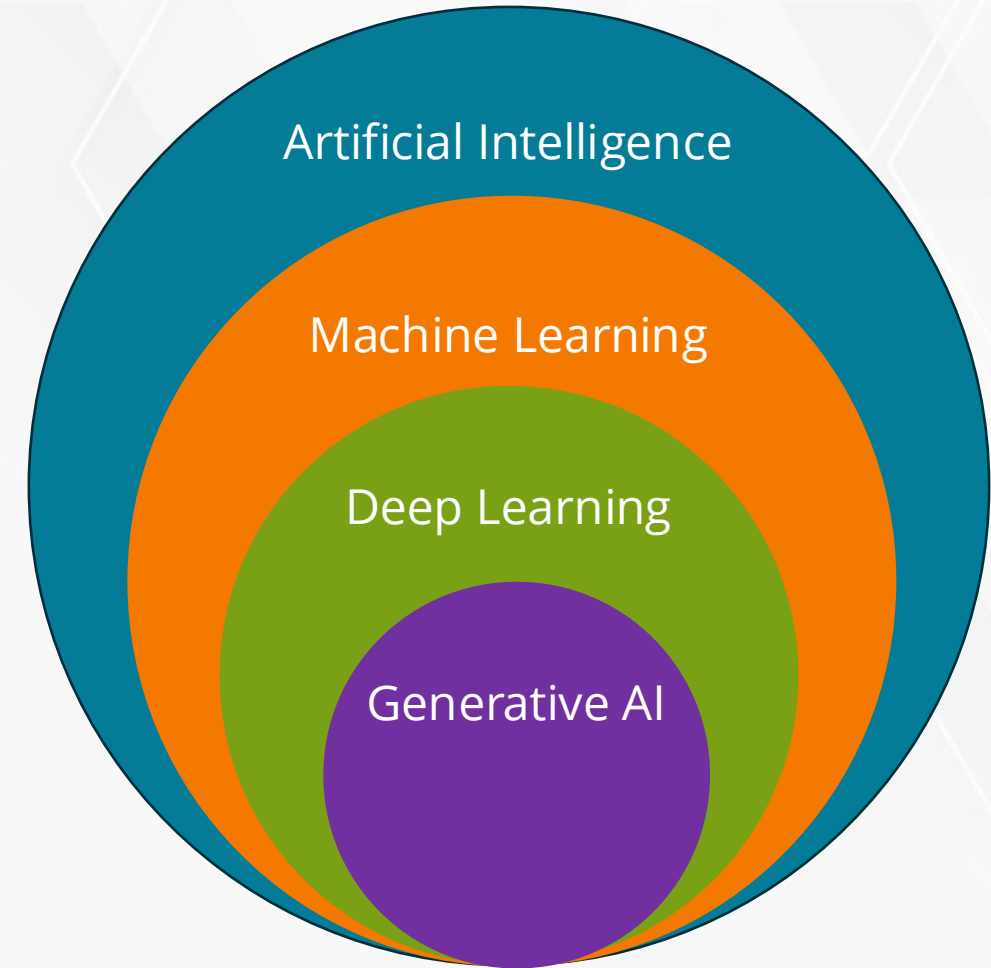
Deep Learning - *Inspired by the Human Brain*

- **Deep Learning** is a special type of Machine Learning that uses **artificial neural networks**, inspired by how the human brain works

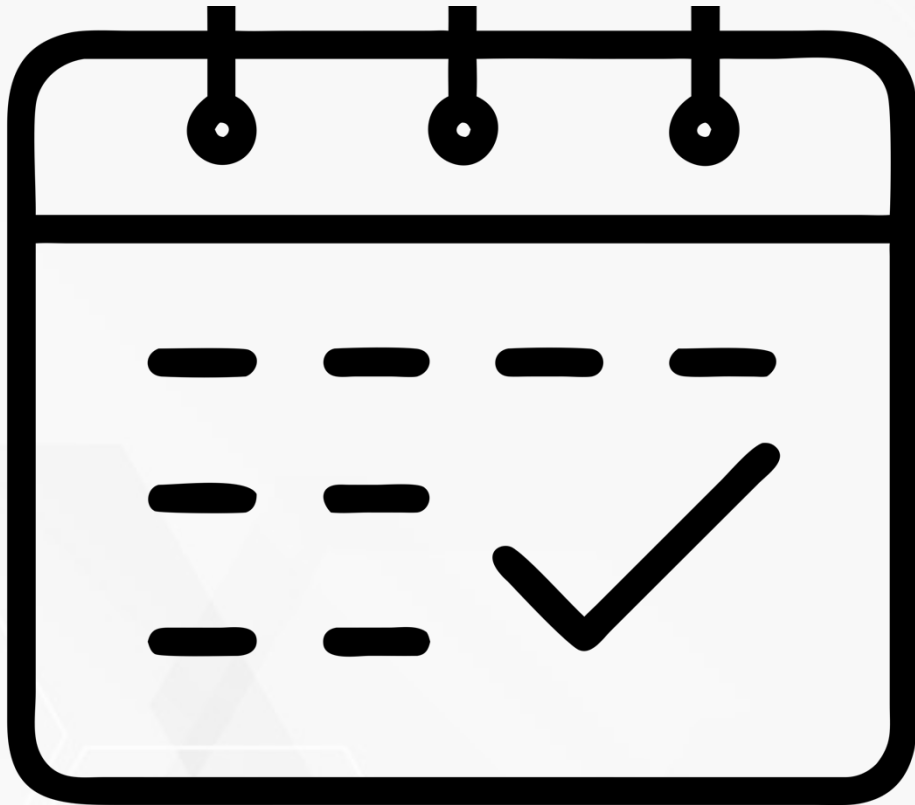


Generative AI

- Unlike traditional AI systems that analyze or predict, **Generative AI creates**. It can write text, generate images, compose music, or even produce computer code.



Real-world use cases



AI - Smart assistants, robotics, autonomous vehicles

ML - Fraud detection, recommendation systems, spam filters

DL - Medical imaging, voice recognition, self-driving systems

GenAI - AI art, text generation, coding assistants, marketing content





What is Machine Learning

Fundamentals of Machine Learning

What is Machine Learning



Examples:

- Predicting exam results
- Recommending learning videos
- Identifying at-risk students

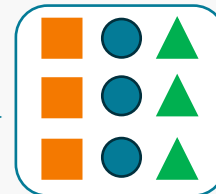
Machine Learning (ML) is a field of Artificial Intelligence where computers learn patterns from data and make predictions or decisions **without being explicitly programmed**.

- Automates decision-making
- Learns from past data using an algorithm
- Improves over time as more data is collected

Features = Input data used for predictions
Example: Study Hours, Attendance



Known Data

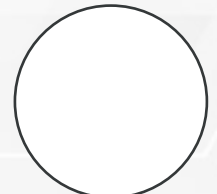


Features



Algorithm

Train



Output

Feature Engineering = Improving and transforming features to help the model learn better

- Convert formats (95% → 0.95)
- Remove missing/wrong data
- Create new helpful features (e.g., Engagement Score = Study Hours + Attendance)

What is Machine Learning



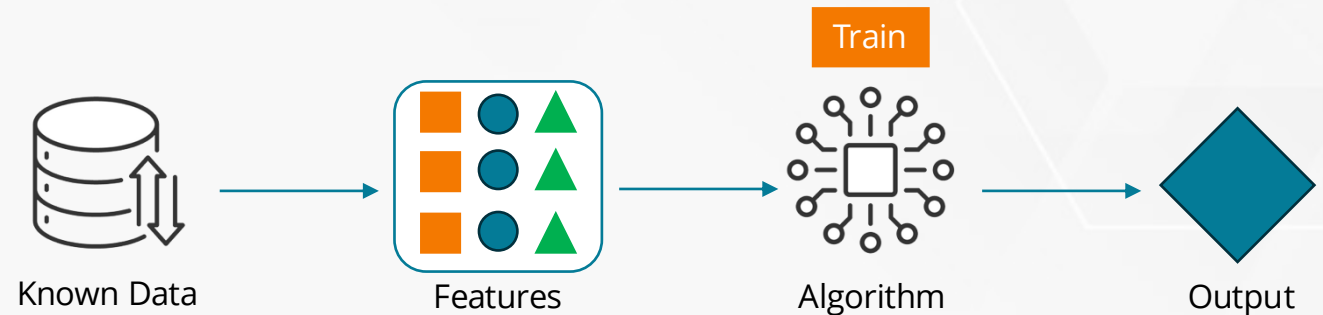
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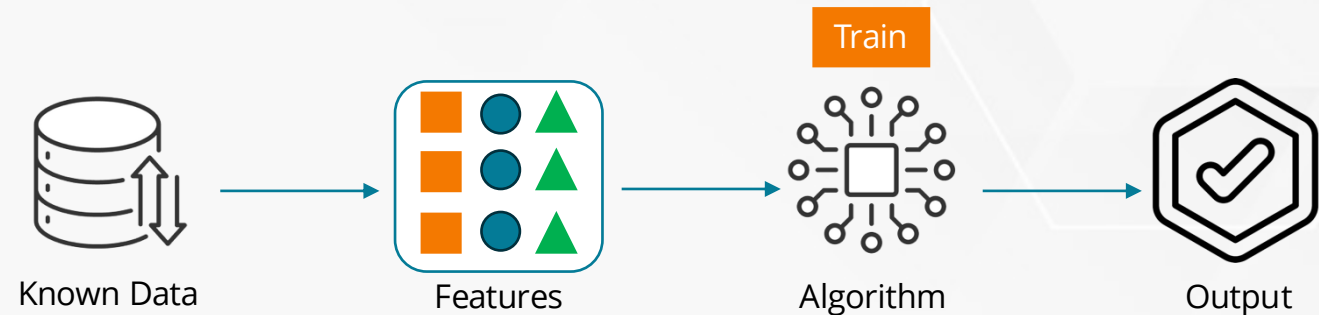
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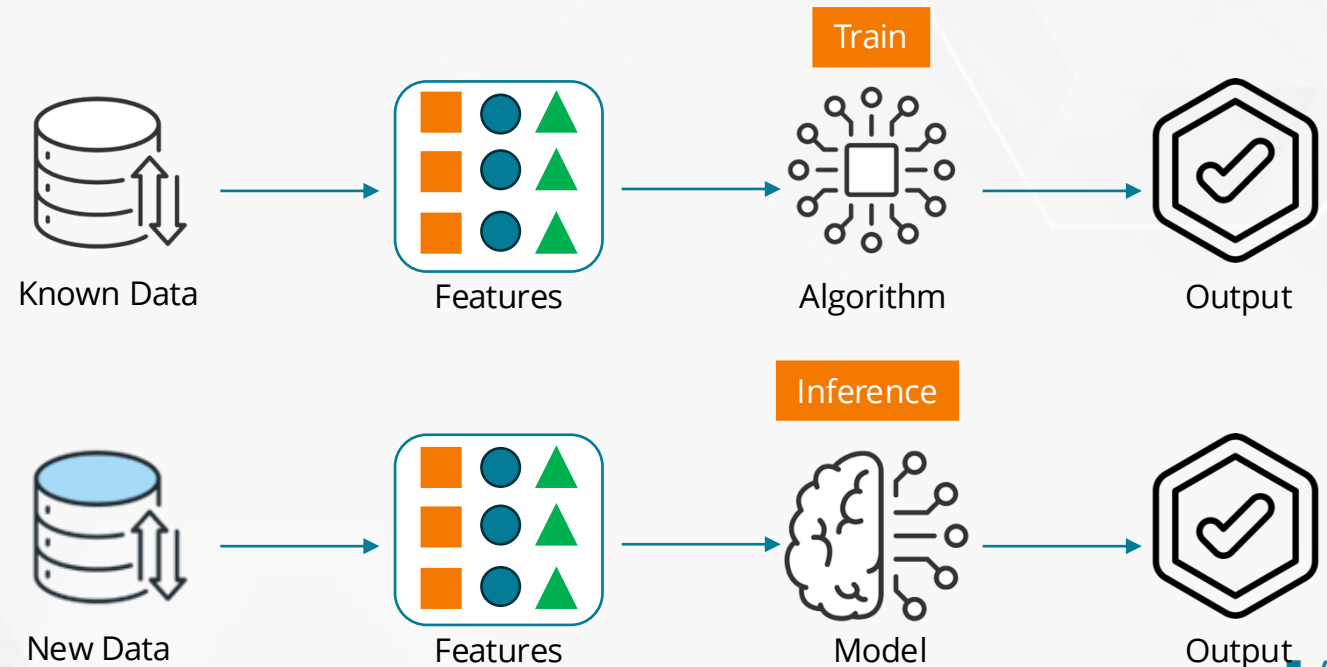
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ML Example – Predicting Student Exam Performance

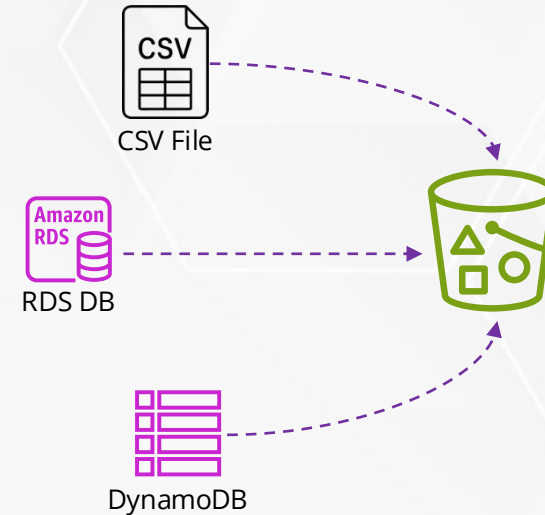
Goal: Predict whether a student will pass or fail based on attendance and study habits.

High-Level Process:

- Collect data → study hours, attendance, past grades
- Process data → clean and prepare for training
- Train model → learn patterns
- Make predictions → for new students

Example Prediction Table:

Student	Study Hours	Attendance	Predicted Result
Anita	10	95%	Pass
Ben	3	70%	Fail
Priya	8	85%	Pass



ML is a tool for learning patterns; not all ML is full AI.



Generative AI

It's cutting-edge technology

What makes Generative AI different?

Traditional ML classifies or predicts. Generative AI creates something new.



Traditional ML

Analyses input and returns a classification or prediction. The answer is always from a fixed set of possible outputs.

EXAMPLES

- Is this email spam? → Yes / No
- Is this image a cat or dog? → Cat
- Will this customer churn? → 73% likely
- What product to recommend? → Product A

Closed-ended — picks from known options



Generative AI

Generates entirely new content that did not exist before. The output is open-ended and unique to the input.

EXAMPLES

- Summarise this 50-page report → [unique summary]
- Write code to sort this list → [new code]
- Translate this document → [translation]
- Answer this question in plain English → [explanation]

Open-ended — creates something new every time

Generative AI: Foundations & How Models Work



Generative vs. Traditional AI

Traditional AI classifies or predicts from existing data. Generative AI creates new, original content that resembles its training data.



Foundation Models

Very large neural networks with billions of parameters, pre-trained on vast data across modalities (language, images). More parameters → more capability.



The Transformer Architecture

Introduced in 'Attention Is All You Need' (2017). Powers modern LLMs like ChatGPT — pre-trained broadly, fine-tunable with little extra data.

How a Model Generates Output

Neural networks + data + prompts work together to predict the next token.

1

Training Data

Vast text, images, code

2

Pre-trained Model

Billions of parameters

3

Prompt + Inference

Input passed at runtime

4

Completion

Next-token predictions

More on transformers

The Transformer Architecture is the core AI design that powers models like ChatGPT, Claude, Gemini, and many modern LLMs.

Simple analogy

- Imagine you're reading a sentence: "The cat sat on the ____"
- Your brain looks at the earlier words ("cat", "sat") to predict the missing word ("mat").

A transformer:

- Reads all the words in a sentence at once (not one-by-one like older AI models)
- Figures out which words are important to each other
- Understands context and relationships
- Predicts the next token (word/part of a word)

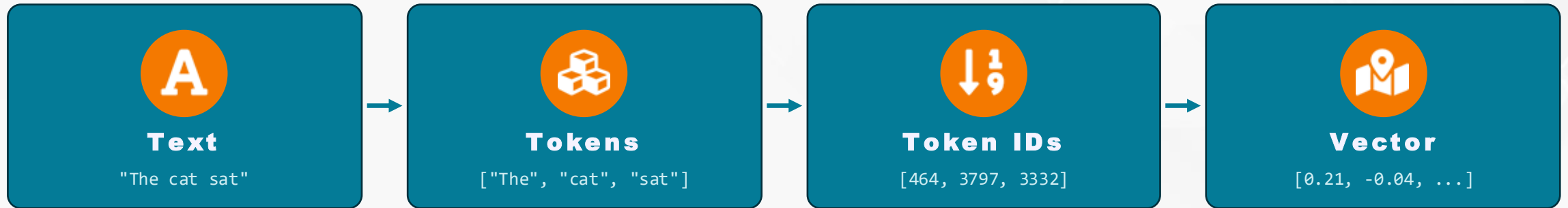


A **transformer** is the neural network architecture that helps GenAI understand relationships between words (or tokens) using attention, enabling intelligent text generation.



From Text to Tokens to Vectors

Models can't read words — only numbers. Here's how text gets translated into something a model can work with.



Tokenizer

The component that splits text into tokens and looks up each token's ID in the model's vocabulary.

A token can be a whole word, part of a word, or even a single character — depending on the tokenizer.

Vector

An ordered list of numbers that pins each token to a specific location in a high-dimensional space.

Think of it like coordinates on a map — the numbers tell the model exactly where this token lives.

Embeddings: Numbers with Meaning

Vectors don't just store tokens — they capture what tokens MEAN, and how meanings relate.



What is an Embedding?

A high-dimensional vector that captures the semantic meaning of a token.

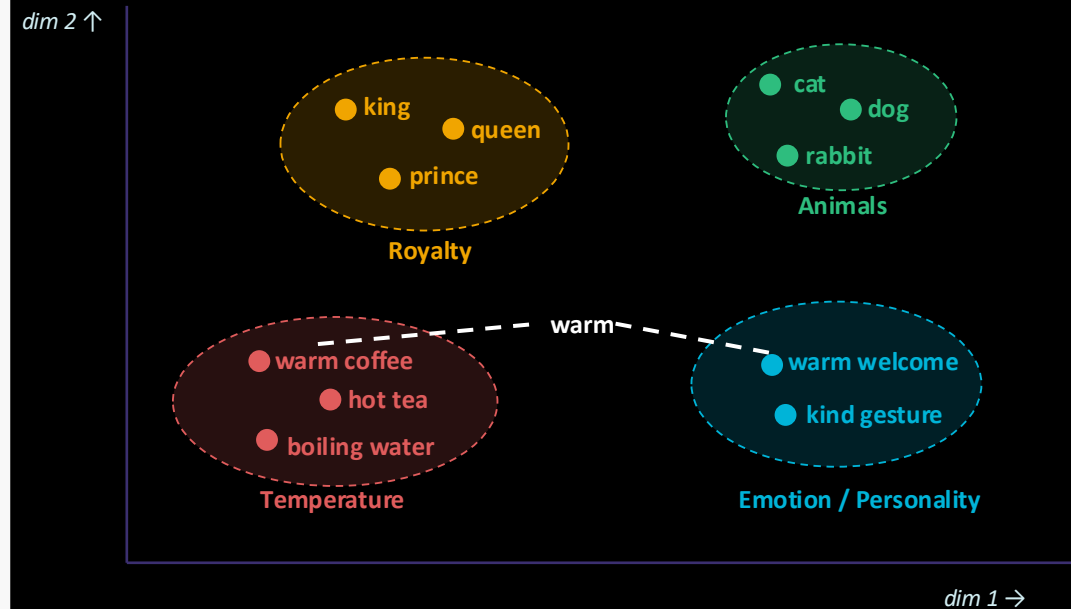
Position = meaning. Tokens that mean similar things sit close together in the vector space.

Relationships are preserved. The distance from "king" to "queen" mirrors the distance from "man" to "woman".

Works for any modality. Text, images, audio, video — all can be embedded.

Semantic Space (illustrative)

Similar tokens cluster together.



Transformers & Self-Attention

The architecture behind every modern LLM — and the trick that lets models understand context.

Self-Attention in Action

Each token decides which OTHER tokens matter when generating output.



Attention weights from "cat":



Key Transformer Concepts



Self-Attention

For each token, computes Query, Key, and Value vectors — then weighs which other tokens matter most.



Long-Range Context

Captures relationships across long sequences — something older RNN architectures struggled with.



Positional Encoding

Tags each token with its position so the model knows word order — "dog bites man" ≠ "man bites dog".



Encoder + Decoder

Stacked layers build up complex representations. Introduced in "Attention Is All You Need" (2017).

How AI reads language

Models can only read a limited amount at once. Long documents need to be split into chunks.

Context Window: The maximum amount of text a model can read in one go. GPT-4 = ~128,000 tokens. But a 500-page report could be millions of tokens — it must be split.

WITHOUT OVERLAP — meaning gets lost at chunk boundaries

Chunk 1: "...The patient was prescribed metformin. The dosage was"

CUT

Chunk 2: "...500mg twice daily. Side effects include nausea..."

✗ "The dosage was" is at the END of Chunk 1 — but "500mg" is at the START of Chunk 2. The model reading Chunk 1 alone never knows the dosage.

WITH OVERLAP — shared tokens bridge the gap

Chunk A: "...prescribed metformin. The dosage was 500mg"

OVERLAP

Chunk B: "The dosage was 500mg twice daily. Side effects..."

✓ Both chunks contain "The dosage was 500mg" — context is preserved. The model reading either chunk understands the full meaning.

Working with Models: Prompts, Inference & Key Terms

Must-Know Vocabulary

The Prompt → Completion Cycle



In-Context Learning

Include examples inside the prompt to guide the model. Three flavors:

Zero-shot — no examples, just the instruction

One-shot — a single example to anchor the task

Few-shot — multiple examples for richer context

Few-shot example

Given the following examples:

Example 1: "Amazon S3 provides highly scalable object storage for files and backups."

Example 2: "Amazon EC2 provides virtual servers for running applications."

Complete the following sentence:

"Amazon RDS provides _____."

Prompt

The input you send to the model.

Inference

Runtime — when the prompt is passed in.

Completion

The model's generated output.

Tokens

Numeric units the model actually processes.

Context Window

Max tokens the model can see at once.

Prompt Engineering

Crafting prompts to get better outputs.



Chain of Thought Prompt



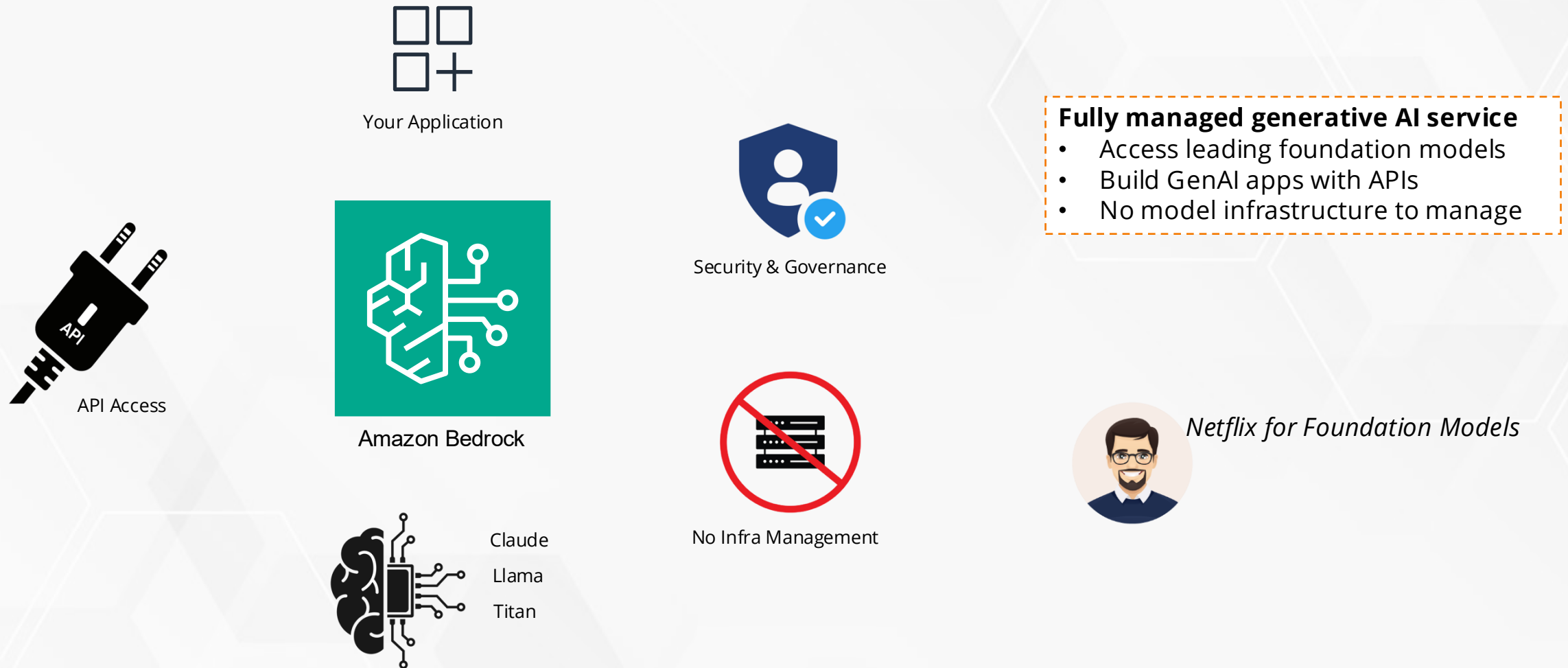
Inference configuration parameters influence how the model responds (we'll discuss these shortly...)



Amazon Bedrock

Access LLMs and build GenAI applications

What is Amazon Bedrock?



Your App → Amazon Bedrock → Foundation Models → AI Response

Not All AI Models Are the Same

	Amazon Nova Micro	Claude Sonnet
Best for	Fast responses	Complex reasoning
Latency	Very low	Higher
Cost profile	Lower	Higher
Ideal use case	Chatbots / quick Q&A	analysis / summarisation / deeper prompts
Provider	Amazon	Anthropic

Same Prompt → Different Answers

Let's compare two models



System prompts make a difference

Inference Configuration Parameters

Control how the model samples its next token at generation time — without changing the model's weights.

TEMPERATURE

Sharpness vs. creativity

Scales the probability distribution before sampling.

- ↓ Low (0.1–0.3): focused, deterministic, repetitive
- ↑ High (0.8–1.2): varied, creative, exploratory

0 = greedy (always top token)

TOP-K

Fixed-size candidate pool

Samples only from the K highest-probability tokens at each step.

- ↓ Small K (e.g. 10): tight, focused output
- ↑ Large K (e.g. 100): broader, more diverse

Ignores everything outside top K

TOP-P

Nucleus sampling

Picks the smallest set of tokens whose cumulative probability $\geq P$.

- ↓ $P = 0.5$: tight nucleus, confident picks
- ↑ $P = 0.9$: wider nucleus, more variety

Adapts to distribution shape

Other inference parameters to know:

Max Tokens

Output length cap

Stop Sequences

Strings that halt generation

Frequency Penalty

Discourages repetition



Introducing RAG

Retrieval Augmented Generation

What is RAG?

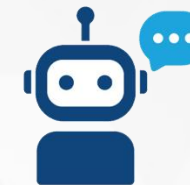
Retrieval Augmented Generation



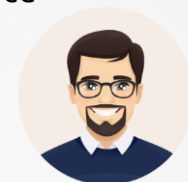
Query corporate data source



Include retrieved data in prompt



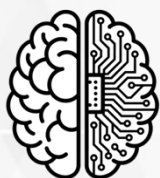
Model creates grounded output



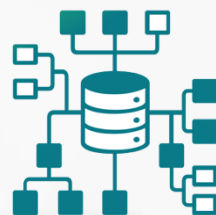
RAG uses semantic search not just keyword search!



Corporate Data



Embeddings Model
Converts data into vectors



Vector Database



RAG Demo

Setting up Bedrock Knowledgebase



Deterministic vs Non-Deterministic

This is one of the most important differences between traditional software and AI-driven workloads.



Traditional Software

DETERMINISTIC

```
sort([3,1,2]) → always [1,2,3]
2 + 2         → always 4
```

Same input → identical output. Every. Single. Time.

- ✓ Unit tests pass reliably
- ✓ Behaviour is predictable and auditable
- ✓ Safe for regulated, high-stakes systems
- ✓ Debugging is straightforward



AI-Driven Workload

NON-DETERMINISTIC

```
"Summarise this report" → response A
"Summarise this report" → response B
"Summarise this report" → response C
```

Same input → similar but never identical output.

- ⚠ Hallucination — confident but wrong answers
- ⚠ Inconsistency — slightly different each time
- ⚠ Bias — reflects patterns in training data
- 🌡 Temperature setting controls randomness



Guardrails

Security for your AI applications

Types of guardrails

Filter Type	Content Filters	Denied Topics	Word Filters	Sensitive Information Filters	Contextual Grounding Checks
Description	Detects harmful content	Blocks entire topics	Blocks exact words or phrases	Deletes or Redacts PII	Reduce hallucinations in RAG



Amazon Rekognition

Cloud-based image and video analysis

Amazon Rekognition



Unsafe Content Detection - Detect and filter explicit, inappropriate, and violent content in images and videos

Amazon **Rekognition** is a cloud-based service that provides **image and video analysis**, allowing developers to add computer vision features to their applications without requiring machine learning expertise. It leverages **deep learning technology** and offers an easy-to-use API that can analyze images and videos stored in Amazon S3.





Amazon Polly

Convert Text to Speech

Amazon Polly



Note: Polly is not a translation service—speech output matches the input language

Amazon Polly Overview

- Converts text into lifelike speech
- Improves accessibility and user engagement
- Supports multiple languages and natural voices

Key Features

- **Input:** Plain text or Speech Synthesis Markup Language (SSML) for advanced control of pronunciation, pitch, speed, and volume
- **Voices:** Multiple male and female options per language; includes bilingual voices (e.g., English + Hindi)
- **Output:** MP3, Ogg Vorbis, PCM formats for web, mobile, IoT, and telephony applications



Amazon Polly



Amazon Textract

Extract Text Data from wide range of
sources

Amazon Textract



Amazon Textract uses machine learning to understand document layouts, not just scan text. This allows it to extract contextually relevant information (like fields in forms or entries in tables) accurately, making the data usable for downstream applications.

Overview

- AI service for extracting text and data from scanned documents
- Detects both typed and handwritten text
- Goes beyond OCR by understanding document structure



Amazon Textract

Key Features

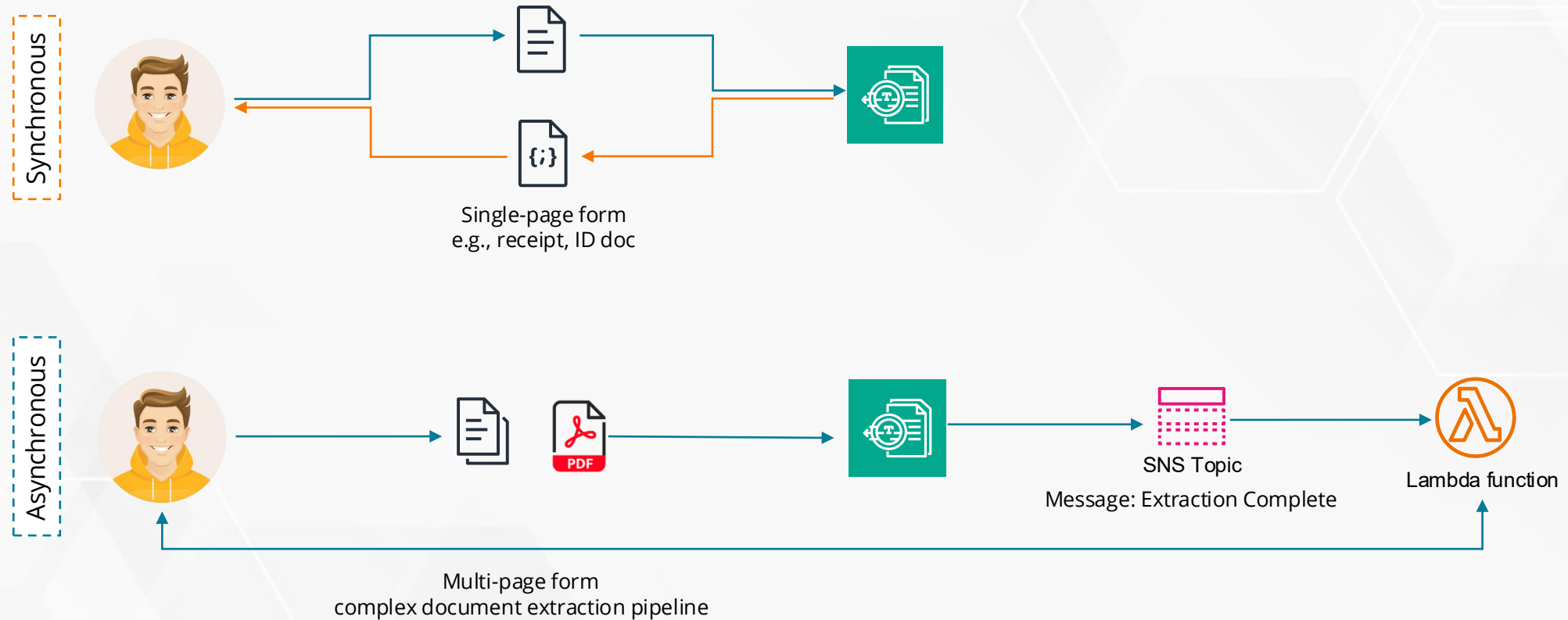
- **Text Extraction:** Identify printed and handwritten text in various documents
- **Forms & Tables:** Extract structured data using the Document Analysis API
- **Queries:** Pull specific information with prebuilt or custom queries
- **Expense Processing:** Extract data from invoices and receipts (AnalyzeExpense API)
- **ID Processing:** Read U.S. driver's licenses and passports (AnalyzeID API)
- **Lending Workflow:** Automate mortgage loan package processing (AnalyzeLending API)

Amazon Textract



Amazon Textract

Synchronous vs Asynchronous Textract Processing





Homework

- Watch the replay
- Think about how you would build a resume generator website using the AWS services we discussed so far in this course.

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Scan any code to connect

